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10 May 1984
Superseding
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3 February 1976

MILITARY SPECIFICATION

VALVE, CONTROL

This specification is approved for use by the Naval Air System Command, Department of the Navy and is available for use by all Departments and Agencies of the Defense Department.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, and test requirements for the Control Valve. The control valve is a solenoid-operated, normally closed valve used in military aircraft to control gas flow from the LAU-7 launcher to the missile for cooling the missile seeker.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, and standards of the issue listed in that issue of Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen Technical

Military

DOD-D-1000	Drawings, Engineering and Associated Lists
MIL-S-4040	Solenoid Electrical, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-B-22020	Bag, Transparent, Flexible, Sealable, Volatile Corrosion Inhibitor, Treated

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MS20813	Cap, Protective

2.1.2 Other Government documents, drawings, and publications. The following drawings form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command

DL 58A164D556	Receiver Assembly, Nitrogen
DL 58A164D863	Receiver Assembly, Nitrogen
DL 58A164D810	Mechanism Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.1.1 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), a first article sample of four control valves shall be delivered to the testing activity designated in the contract or

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purchase order. The first article sample shall be manufactured using the same methods, materials, processes and procedures proposed for production. Any production prior to acceptance of the first article sample is at the contractor's risk.

3.2 Selection and quality verification of parts and materials. Quality verified parts (see 3.2.1) and materials from approved sources (see 3.2.2) shall be used in the fabrication of the end item. The contractor shall prepare test reports of conformance to applicable requirements and make them available for the procuring activity (see 6.2.2).

3.2.1 Approval of parts and materials. Parts and materials proposed for use in the fabrication of the end item that are not specified in detail herein or in applicable referenced documents shall be selected from other documents in the order of precedence specified in MIL-STD-143 and shall be approved by the procuring activity prior to use (see 6.2.2). Approval shall be predicated upon source (vendor) approval and verification of part quality by test.

3.2.2 Source approval. The contractor shall be responsible for ascertaining that each proposed source for parts and materials maintains a quality control program to ensure consistent delivery of products conforming to applicable requirements. When parts or materials are fabricated by the contractor, the area used for that purpose shall be considered the same as an outside vendor insofar as source approval is concerned. Source approvals by the contractor shall be subject to concurrence by the procuring activity. Source approval will be granted by the procuring activity upon preparation of acceptable evidence of any of the following (see 6.2.2):

a. Formal on-site survey of the vendor's manufacturing facilities.

b. The source has been previously approved, and evidence is available to confirm that acceptable product quality has been maintained.

c. Other evidence of acceptable product quality is available.

3.2.3 Verification of part quality. Quality verification shall be demonstrated by successful testing of the part to levels of environmental, electrical, and physical stresses equal to or greater than those which will be encountered in the system and as required by the part specification. Part quality may be demonstrated by any of the following:

a. When the part is manufactured and qualified to a military specification which includes requirements equal to or more severe than those of the end item application.

b. When the part is manufactured and qualified to a military specification not including all requirements of the application, and additional tests for such requirements have been performed.

c. When the part is not qualified to a military specification, but all tests appropriate to the application have been performed.

3.2.4 Approved source/quality verified parts list (AS/QVPL). An AS/QVPL shall be prepared by the contractor for all subassemblies and parts and made

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available to the procuring activity for approval (see 6.2.2). All entries on the AS/QVPL shall be fully substantiated by the preparation of data to satisfy the requirements of source approval and part test approval as specified herein. Reports of all quality verification testing shall conform to the requirements of MIL-STD-831. The contractor may solicit from the procuring activity information as to the status of quality verified parts and approved sources which may have been previously developed. Neither part environmental tests nor part environmental test reports are required in those instances where vendors are selected from current AS/QVPLs of other programs. Entries on the AS/QVPL shall identify the vendor and the product which is proposed for use in the control valve by the contractor. In the event several sources have previously been approved for the application, the contractor shall specify the source of the parts to be used. Each item listed shall be supported in accordance with the provisions of this section. This listing does not supplant or modify the applicable documentation or other Government Qualified Parts Listing.

3.2.4.1 Conditions governing the quality verification of parts. When the contractor proposes to demonstrate the quality verification of parts, as provided herein, the following provisions shall apply:

a. Test may be performed by the contractor, utilizing his own facilities, the facility of the manufacturer of the part or by any other suitable facility selected by the contractor. In any event, the facilities shall be approved by the contractor prior to the performance of any test. This information shall be made available to the procuring activity for approval (see 6.2.2).

b. Prior to the performance of any test, a detailed test procedure shall be prepared by the contractor and made available to the procuring activity for approval (see 6.2.2). The procedure shall identify, as a minimum, the step by step functions to be performed and the test equipment to be used, and shall provide for full conformance with the provisions of the applicable documentation. The Government reserves the right to review any of the individual test procedures at its discretion.

c. Quality verification testing may be monitored by a representative of the procuring activity.

3.2.4.2 Changes to the AS/QVPL. The contractor may develop alternative sources for any part, subject to the provisions herein. Proposals by the contractor for part substitution or changes to the AS/QVPL shall include proof that the proposed substitute part will meet the requirements of 3.2.2 and 3.2.3. The AS/QVPL shall be revised and updated as applicable to include approved changes (see 6.2.2).

3.2.4.3 Removal from the AS/QVPL. Evidence of poor quality may be cause for suspension or removal of a source from the AS/QVPL at the option of the contractor or the procuring activity (see 6.2.1). The contractor shall develop quality verified sources for those parts for which the approved source has been removed. Quality reverification shall be required under any of the following conditions unless otherwise directed by the procuring activity:

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a. When the part has been modified by a change in design, materials, manufacturing process, manufacturing inspection procedures, or manufacturing facilities. The procuring activity shall be notified prior to delivery of any parts incorporating any such changes and be provided quantitative evidence of the effect of such changes on the performance or other characteristics of the control valve.

b. When there is evidence that the quality of the part has not been maintained. This evidence may be in the form of accumulated failure reports which indicated that the quality of the part has deterioriated, or there is a failure trend.

c. When the manufacturer has not produced the item for six or more months.

3.2.5 Part conformance provisions. When a drawing or specification for a part invokes by reference the provisions of a military specification, total conformance to the provisions of the referenced specification shall be required. Total conformance is defined as conformance to the stated performance and product characteristics, and compliance with all quality assurance provisions, including qualification, quality verification, and quality conformance testing.

3.3 Design and construction. The design and construction of the control valve shall be in accordance with MIL-S-4040 and with DL 58A164D810 and as specified herein. The contractor shall notify the procuring activity of any changes in design and construction of the control valve which occur after it has been accepted by the procuring activity (see 6.2.2).

3.3.1 Materials. Materials shall be selected on the basis of strength, workability and compatibility with other materials in the item so as to minimize galvanic action, corrosion, etc. Materials used in manufacturing the solenoid shall be in accordance with MIL-S-4040.

3.3.2 Finishes. Finishes shall be applied to all materials, whenever applicable and shall be for the purpose intended, such as corrosion resistance.

3.3.3 Filter. The control valve shall be provided with a filter which will prevent particles 5 microns or larger from entering the assembly.

3.3.4 Weight. The weight of the control valves shall be not greater than two pounds.

3.3.5 Interchangeability. All control valves having the same part number shall be interchangeable. Change in part numbers shall be governed by drawing number requirements of DOD-D-1000.

3.3.6 Identification of parts. All control valves shall be marked as specified in DL 58A164D810.

3.3.7 Dimensions. The control valve envelope dimensions shall be in accordance with the applicable drawing for the control valve being procured.

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3.4 Construction. The control valves, when constructed in accordance with DL 58A164D810, shall meet the requirements specified in 3.4.1 through 3.4.7.

3.4.1 Power requirement. The control valve shall perform as required under all applicable conditions specified herein from a 28 VDC source except that the control valve shall be capable of operating over a range of 17 to 30 VDC measured at the solenoid input. The current drain shall be not greater than one ampere at 28 VDC and 25 ± 5° C ambient.

3.4.2 Pressure requirements. The control valve shall be capable of specified operation at any pressure from 15 to 3800 psig.

3.4.3 Operating time. The solenoid shall open or close the valve in less than one second after application of 17 VDC.

3.4.4 Pressure drop. The pressure drop through the valve shall be not greater than 100 psig with an applied pressure of 2800 psig and a flow rate of 28.3 liters per minute (one cubic foot per minute).

3.4.5 Adaptability. The control valve shall be capable of functioning with 2° misalignment with the seal valve, part number 58A164D859, axis. Misalignment shall be in any direction.

3.4.6 Leakage. The control valve shall not leak in excess of two cubic centimeters per minute when subjected to input pressures between 200 and 3800 pounds per square inch gage (psig) with the solenoid deenergized.

3.4.7 Attitude. The control valve shall be capable of functioning in any attitude.

3.5 Cleanliness. The control valve shall be free of all water, oil, grease, and foreign matter exceeding 0.002 inch in size on all surfaces.

3.6 Proof pressure. The control valve shall not be damaged and shall operate as specified after being subjected to 5000 psig.

3.7 Burst pressure. The control valve shall not be deformed or show physical damage after being subjected to a burst pressure of 6700 psig for a duration of at least 30 minutes.

3.8 Environmental. The control valve shall meet the following environmental requirements of MIL-STD-810 unless otherwise specified herein.

3.8.1 High temperature. The control valve shall operate as required during and after the test of MIL-STD-810, Method 501, Procedure I, except the temperature shall be 121°C (250° F) (see 4.6.8.1).

3.8.2 Low temperature. The control valve shall operate as required during and after the test of MIL-STD-810, Method 502, Procedure I for a period of four hours (see 4.6.8.2).

3.8.3 Temperature shock. The control valve shall operate as required after the test of MIL-STD-810, Method 3, Procedure I (see 4.6.8.3).

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3.8.4 Dust. The control valve shall operate as required after the test of MIL-STD-810, Method 510, Procedure I (see 4.6.8.4).

3.8.5 Salt fog. The control valve shall operate as required after the test of MIL-STD-810, Method 509, Procedure I, except the duration shall be 50 hours (see 4.6.8.5).

3.8.6 Humidity. The control valve shall operate as required after the test of MIL-STD-810, Method 507, Procedure I, except four 24 hour cycles shall be substituted for the ten 24 hour cycles (see 4.6.8.6).

3.8.7 Vibration. The control valve shall operate as required during and after exposure to the vibration tests specified herein (see 4.6.8.7).

3.8.8 Shock. The control valve shall operate as required after the test of MIL-STD-810, Method 516, Figure 516-2, except the acceleration shall be 25 g and shall reach peak value in not more than 0.600 second and not less than 0.015 second (see 4.6.11).

3.9 Service life. The control valve shall operate as required and show no damage after being subjected to cycling tests as specified herein. (see 4.6.9).

3.10 Precedence. The selection of specifications and standards for use in the design and development shall be in accordance with the requirements of MIL-STD-143. When the requirements, this specification, and referenced documents are in conflict, the following precedence shall apply:

a. This specification. This specification shall have precedence over all referenced documents.

b. Referenced documents. Any documents referenced in this specification shall have precedence over all subsidiary documents.

3.11 Workmanship. The workmanship displayed in fabrication and assembly of the control valve shall be such as to assure, within design limitations, the ability of the control valve to meet its performance requirements under all conditions specified herein. Particular attention shall be paid to neatness and thoroughness of assembly, marking of parts and assembly, plating, painting, machining, welding, brazing and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in the approved first article sample, subject to any qualification stated in the Government's notice of approval, shall determine the requirements of the remainder of the contract relative to workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements except as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in

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the specification where such inspections are deemed necessary to ensure that the supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the following conditions and tolerances.

a. Room temperature. Room ambient of 25 ± 10°C (77 ±18°F) indicated.

b. Test temperature. All parts of the test item shall be stabilized at the specified temperature ±2.8°C (±5°F) indicated prior to conducting any operational tests. Unless otherwise specified herein, temperature stabilization will have been attained when the indicated temperature of the surface of the largest mass of the test item does not change more than ±2.8°C (±5°F) in a period of one hour.

c. Humidity. Room ambient to 95 percent relative humidity.

d. Test gas. The test gas shall be nitrogen in accordance with BB-N-411, Type I, Class I, Grade A.

e. Test fixture. A seal valve depicted in DL 58A164D863, DL 58A164D556, or an equivalent fixture and adapter shall be used in mating the inlet probe of the control valve. Tighten seal valve with adapter to a torque of 100 inch pounds. The seal valve and adapter shall be used in all tests except shock. The test set up shown in figure 1 shall be used for shock and vibration testing.

f. At the conclusion of any test requiring exposure to high temperature, the control valve shall be allowed to cool to room ambient prior to release of the inlet pressure. Inlet pressure shall be released gradually during a 15 minute interval.

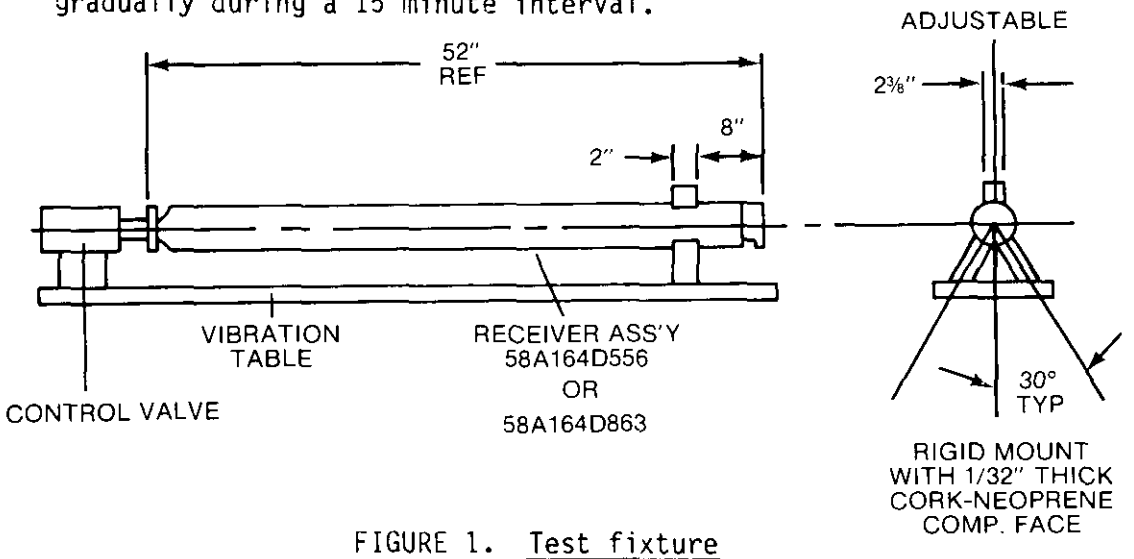


FIGURE 1. Test fixture

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4.4 First article inspection. First article inspection shall consist of the tests specified in Table I. The procuring activity reserves the right to delete tests and alter the test sequence (see 6.2.1). First article approval is only valid on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.4.1 Control valve samples. First article control valve samples shall consist of the first four control valves manufactured on the contract or purchase order. The control valves shall be permanently labeled for ease of identification and shall be produced with equipment and procedures that will be used for the production of all the control valves designated in the contract or purchase order. Prior to submission, the contractor shall inspect and test the samples to the degree necessary to assure that they conform to the requirements of the contract. Control valves shall be forwarded to the designated test facility specified in the contract (see 6.2.1) for testing. Approval of the first article sample shall be by the procuring activity upon satisfactory completion of all first article tests and a test report being made available to the procuring activity (see 6.2.2).

4.5 Quality conformance inspections. Acceptance of control valves shall be on an inspection lot basis. Quality conformance inspections shall consist of the following tests:

a. Individual tests (see Table II, Group 1).

b. Sampling tests (see Table II, Groups 1 and 2).

4.5.1 Individual tests. Individual tests are those tests conducted by the contractor on each control valve submitted for acceptance under the contract or purchase order. Individual tests for control valves are shown in Table II, Group 1. Failure to pass these tests shall be cause for rejection.

4.5.2 Sampling tests. A random sampling of control valves shall be selected by the procuring activity from each inspection lot in accordance with Table III and shall be subjected to the tests listed in Table II, Groups 1 and 2. The sample valves shall be forwarded to the designated testing activity specified in the contract or purchase order. (See 6.2.1 for sample testing.) Group 1 tests shall be conducted prior to Group 2 tests. Failure to pass these tests shall be cause for rejection of the lot. The procuring activity reserves the right to delete tests and alter the test sequence of Table II (see 6.2.1). In the event any sample representing an inspection lot fails to meet the inspection and test requirements outlined herein, the contractor, at the discretion of the procuring activity, may have the opportunity of reworking the entire affected lot. The procuring activity shall determine which inspections and tests of this specification are to be performed on the reworked inspection lot samples. The number of test samples shall be at the discretion of the procuring activity (see 6.2.1).

4.6 Inspection methods.

4.6.1 Cleanliness. The control valve shall be subjected to the cleanliness test detailed herein. The test shall be conducted in a clean, dust-free atmosphere and all equipment shall be thoroughly cleaned. The control valve

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TABLE I. First article inspections - schedule and sequence 1/

Test	Test Title	Test Para.	Sample			
			A	B	C	D
A	Cleanliness	4.6.1	1	1	1	1
B	Inspection	4.6.2	2	2	2	2
D	Dielectric Strength	4.6.6	3	3	3	3
E	Insulation Resistance	4.6.7	4	4	4	4
F	Proof Pressure	4.6.4	5	5	5	5
G	Leakage	4.6.5	6	6	6	6
H	Normal Operation	4.6.3.1	7	7	7	7
J	Pressure Drop	4.6.3.2	8	8	8	8
K	High Temperature	4.6.8.1	9	X	9	X
L	Low Temperature	4.6.8.2	10	X	10	X
M	Thermal	4.6.8.3	X	9	X	9
N	Dust	4.6.8.4	X	10	X	
P	Salt Fog	4.6.8.5	X	11	X	10
Q	Humidity	4.6.8.6	X	12	X	11
R	Vibration	4.6.8.7	11	X	11	X
S	Shock	4.6.11	12	X	12	X
T	Life	4.6.9	X	13	X	12
U	Burst	4.6.10	13	X	13	X

1/ Suggested sequence is denoted by the number in the sample column. An X denotes test is not required. Test sequence may vary to facilitate testing.

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TABLE II. Quality conformance inspections - schedule and sequence 1/

Tests	Test Para.	Sample
Group 1/		All Samples
Cleanliness	4.6.1	1
Inspection	4.6.2	2
Dielectric Strength	4.6.6	3
Insulation Resistance	4.6.7	4
Proof Pressure	4.6.4	5
Leakage	4.6.5	6
Functional Test	4.6.3	7
Group 2/		A B C D
High Temperature	4.6.8.1	1 X 1 X
Low Temperature	4.6.8.2	2 X 2 X
Temperature Shock	4.6.8.3	X 1 X 1
Dust	4.6.8.4	X 2 X 2
Salt Fog	4.6.8.5	X 3 X 3
Humidity	4.6.8.6	X 4 X 4
Vibration	4.6.8.7	3 X 3 X
Shock	4.6.11	4 X 4 X
Life	4.6.9	X 5 X 5
Burst	4.6.10	5 X 5 X

1/ Test sequence is denoted by the number in the sample columns. An X denotes test is not required.

2/ The test sequence is the suggested test sequence. Test sequence may vary at the discretion of the test facility.

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shall be visually examined for cleanliness and shall be free of all water, oils, grease and foreign matter exceeding 0.002 inch in size on all surfaces internally interfaced with the nitrogen receiver.

4.6.2 Inspection. The control valve shall be inspected as specified in 4.6.2.1 and 4.6.2.2.

4.6.2.1 Individual tests-visual. The control valve shall be inspected to ensure that it has been properly assembled, and that the workmanship is as specified in 3.11.

4.6.2.2 First article and quality conformance tests.

a. The control valve shall be thoroughly examined to determine conformance to all the requirements of the control valve drawing listed on DL 58A164D810.

b. The control valve shall be examined to ascertain that the packaging, packing, and marking are in accordance with Section 5.

4.6.3 Functional tests. The control valve shall be subjected to the functional tests specified in 4.6.3.1 and 4.6.3.2.

TABLE III. Sampling test items.

Lot Size	Number of samples to be tested
100 or less	2
101 to 300	3
301 to 500	4
501 or more	4 plus 1 for each additional 500 or fraction thereof.

4.6.3.1 Normal operation. To perform a normal operation test requires the control valve to be mated to a seal valve assembly (part number 58A164D859) with adapter or an equivalent test set-up. The adapter is connected to a source of nitrogen at 2800 (+100,-000) psig. The output fitting is connected to a flow meter. Application of 17 (+1,-0) VDC to the control valve solenoid leads shall cause the valve to actuate and deliver at least 28.3 liters per minute (1 cubic foot per minute). The current input to the solenoid shall be not greater than 0.9 ampere and the operating time shall be less than one second. The test shall be repeated at an actuation voltage of 30 (+0,-1) VDC.

4.6.3.2 Pressure drop. The pressure drop across the valve, while subjected to a flow rate of 28.3 liters per minute (1 cubic foot per minute) at an inlet pressure of 3000 ±100 psig, shall be not greater than 100 psig.

4.6.4 Proof pressure. The control valve shall be subjected to a proof pressure test under the following conditions:

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a. Inlet probe pressurized to 5500 ±100 psig for three minutes with the valve closed (solenoid deenergized).

b. Inlet probe pressurized to 5500 ±100 psig for three minutes and the control valve opened (solenoid energized).

4.6.5 Leakage. The control valve shall be checked for leakage at the following pressures:

a. 200 psig
b. 1000 psig
c. 2200 psig
d. 3000 psig
e. 3800 psig

The leakage test shall be performed under the following conditions:

a. Test set shall be in accordance with figures 2 and 3.

b. Control valve closed (solenoid deenergized) and outlet fitting open to atmospheric pressure. Leakage shall be not greater than 2 cubic centimeters per minute.

c. The control valve open (solenoid energized) and the outlet fitting sealed. Leakage shall be not greater than one cubic centimeter per minute.

4.6.5.1 Leakage at temperature extremes. Leakage at -20°C and +121°C shall be not greater than values listed in 4.6.5.

4.6.6 Dielectric strength. The control valve solenoid shall be tested in accordance with Method 301 of MIL-STD-202. The following details shall apply:

a. Test voltage 1000 ±50 VAC, 60 ±10Hz.

b. Apply test voltage between solenoid lead and control valve body.

c. Current flow shall be not greater than 1.0 milliampere during the test period.

d. The control valve shall meet the normal operating requirements of 4.6.3.1 and 4.6.3.2 at the conclusion of the test.

4.6.7 Insulation resistance. Insulation resistance of the control valve solenoid coil shall be measured in accordance with MIL-STD-202. The following details shall apply:

a. Test condition A

b. Test temperature 85°C

The control valve shall meet the normal operating requirements of 4.6.3.1 and 4.6.3.2 at the conclusion of the test after being allowed to cool at room ambient temperature for two or more hours.

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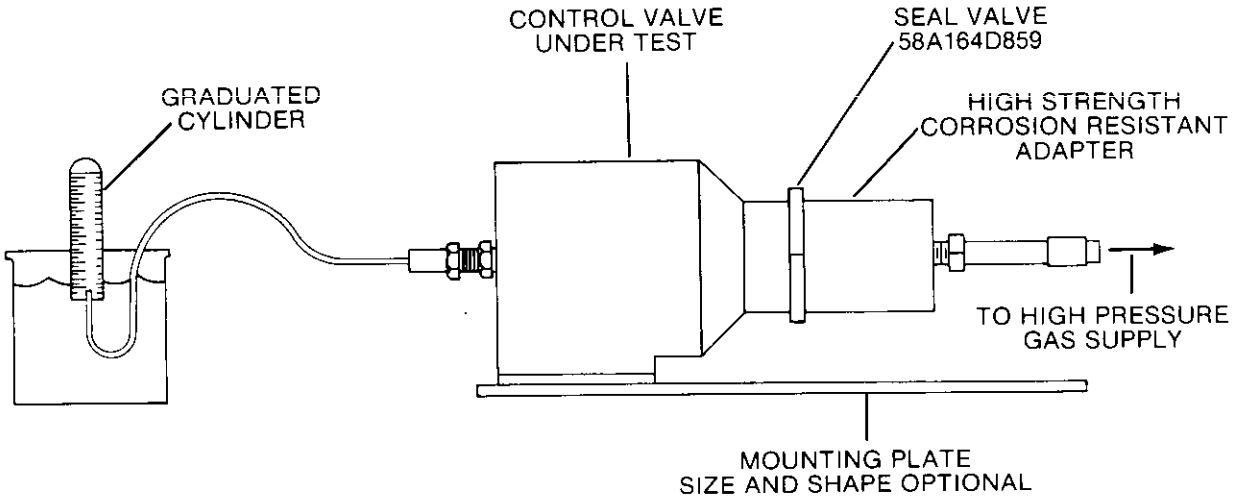


FIGURE 2. Test set-up for measuring leakage through control valve

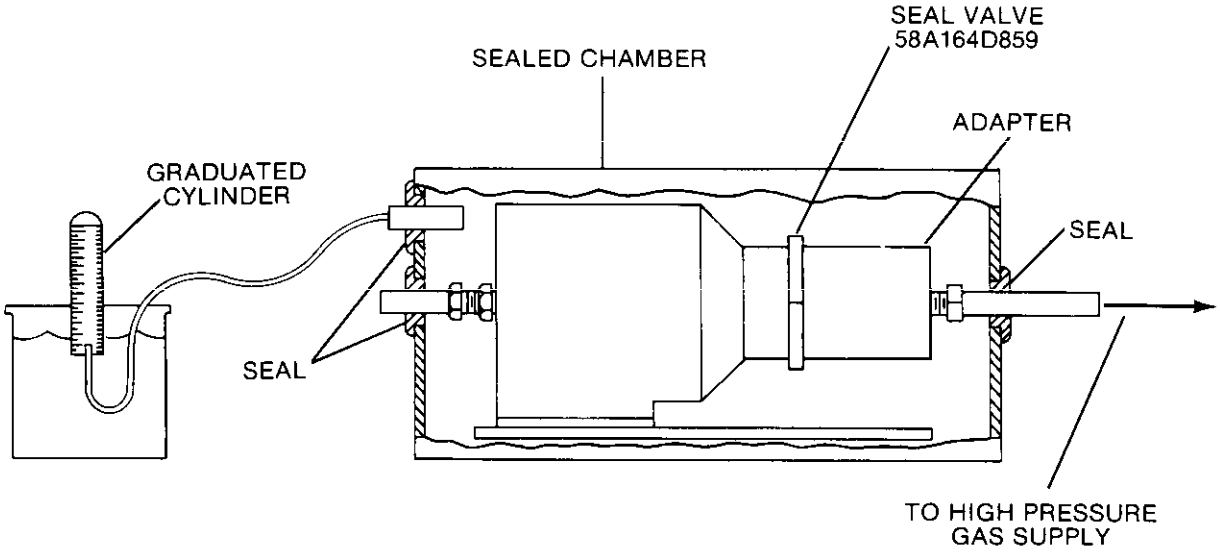


FIGURE 3. Test set-up for measuring external leakage

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4.6.8 Environmental tests. The control valve shall be subjected to the environmental tests specified in 4.6.8.1 through 4.6.8.6.

4.6.8.1 High temperature. The control valve shall be subjected to the high temperature test in accordance with MIL-STD-810, Method 501, Procedure I, except the temperature shall be 121°C (250°F). While at temperature conditions, the control valve shall be closed and the inlet probe pressurized to and maintained at 3800 (+0,-200) psig. At the completion of the test, the control valve shall be visually inspected for damage and tested in accordance with 4.6.3 and 4.6.5.

4.6.8.2 Low temperature. The control valve shall be subjected to the low temperature test in accordance with MIL-STD-810, Method 502, Procedure I for a period of four hours. The storage temperature shall be -45°C (-65°F). While at temperature conditions, the control valve shall be closed, and the inlet probe pressurized and maintained at 3800 ±100 psig. At the completion of the test, the control valve shall be visually inspected for damage and tested in accordance with 4.6.3 and 4.6.5.

4.6.8.3 Thermal shock. The control valve shall be subjected to the thermal shock test in accordance with MIL-STD-810, Method 503, Procedure I. Prior to being subjected to the temperature conditions, the control valve shall be closed and the inlet probe pressurized to 3000 ±100 psig. At the completion of the test, the control valve shall be visually inspected for damage and tested in accordance with 4.6.3 and 4.6.5.

4.6.8.4. Dust. The control valve shall be subjected to the dust test in accordance with MIL-STD-810, Method 510, Procedure I. While at test conditions, the control valve shall be closed and the inlet probe pressurized to 225 ±25 psig. The outlet fittings shall be capped. At the completion of testing, the control valve shall be visually inspected for damage and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.8.5 Salt fog. The control valve shall be subjected to the salt fog test in accordance with MIL-STD-810, Method 509, Procedure I except the duration shall be 50 hours. While at test conditions, the control valve shall be closed and the inlet probe pressurized to 225 ±25 psig. The outlet fitting shall be capped. At the completion of testing, the control valve shall be visually inspected for damage and corrosion, and tested in accordance with 4.6.3.

4.6.8.6 Humidity. The control valve shall be subjected to the humidity test in accordance with MIL-STD-810, Method 507, Procedure I, except four 24-hour cycles shall be substituted for the ten 24-hour cycles. While at test conditions, the control valve shall be closed and the inlet probe pressurized to 225 ±25 psig. The outlet fitting shall be capped. At the completion of testing, the control valve shall be visually inspected for damage and corrosion, and tested in accordance with 4.6.3.2 and 4.6.3.3.

4.6.8.7 Vibration. The control valve shall be subjected to the tests specified in 4.6.8.7.1 through 4.6.8.7.5.

4.6.8.7.1 General. The vibration tests specified herein shall be performed in each of three mutually perpendicular axes in the order shown. All testing

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shall be performed in one axis prior to testing in the next axis. As indicated herein, the control valve shall be checked for proper operation.

4.6.8.7.2 Control valve operation. While subjected to vibration, the control valve shall be closed and the inlet probe pressurized 3000 ±100 psig. The output fitting shall be capped. After each axis of vibration is completed, the control valve shall be visually inspected for damage and tested in accordance with 4.6.3. and 4.6.3.3.

4.6.8.7.3 Mounting techniques. The control valve shall be attached to the vibration exciter table by means of the fixture specified in 4.3.e. The vibration input shall be measured on the fixture at a point adjacent to the mounting point of the control valve.

4.6.8.7.4 Sinusoidal test level. The control valve shall be subjected to sinusoidal vibration along each of three mutually perpendicular axes. The frequency shall be ±2 percent and the amplitude shall be ±5 percent. The test shall consist of sweeping for 60 minutes in each axis from 5 to 2000 Hz, at one octave per minute in accordance with Table IV.

TABLE IV. Sinusoidal test level.

Frequency	Displacement or acceleration
5 - 20 Hz	0.100 inch da
20 - 33 Hz	±2.0 g
33 - 74 Hz	0.036 inch da
74 - 2000 Hz	±10.0

4.6.8.7.5 Random test level. The control valve shall be subjected to a random vibration along each of three mutually perpendicular axes. The instantaneous random vibration acceleration peaks shall be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than +100, -50 percent (±3 dB) between 20 Hz and 2,000 Hz, except that deviations as large as +300, -75 percent (±6 dB) shall be allowed over a cumulative bandwidth of 100 Hz maximum between 500 and 2,000 Hz.

Tolerance levels in terms of dB are defined as:

$$dB = 10 \log \frac{W_1}{W_0}$$

Where W_1 = measured acceleration power spectral density of g^2/Hz units.

The term W_0 defines the specified level in g^2/Hz units.

Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant

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product. BT = 50 minimum. Specific analyzer characteristics shall be as specified below or equivalent.

a. On-line, contiguous filter, equalization/analysis system having a bandwidth = B = 50 Hz maximum.

b. Swept frequency analysis systems characterized as follows:

(1) Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B = 20 Hz, maximum between 20 and 200 Hz.

B = 50 Hz, maximum between 200 and 2,000 Hz.

(b) Analyzer averaging time = T = 2 RC = 1 second minimum, where T = true averaging time and RC = analyzer time constant.

(c) Analysis sweep rate (linear) = $S = \frac{B}{4RC}$ or $\frac{B^2}{8}$ Hz/sec, whichever is smaller.

(2) Constant percentage bandwidth analyzer.

(a) Filter bandwidth = pf_c = one third octave minimum ($0.23f_c$) where p = percentage and f_c = analyzer center frequency.

(b) Analyzer averaging time = $T = \frac{50}{pf_c}$, minimum.

(c) Analysis sweep rate (logarithmic) = $S = \frac{pf_c}{4RC}$, or $\frac{(pf_c)^2}{8}$ Hz/second, whichever is smaller.

c. Digital-power, spectral-density analysis systems employing quantization techniques shall provide accuracies corresponding to the above approach.

The random vibration test level shall be a flat spectrum in the range of 20 to 2000 Hz at a PSD level of 0.02 g^2/Hz with an overall test level of 6.3 g's rms. The test time shall be 40 minutes in each of the three mutually perpendicular axes. The analysis shall be as shown in Table V.

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TABLE V. Vibration analysis.

Bandwidth	Analysis filter
20 - 200 Hz	20 Hz filter
200 - 2000 Hz	50 Hz filter

The analysis shall yield a statistical accuracy of 120 degrees of freedom.

4.6.9 Service life. The control valve shall be subjected to a 5000 cycle service life test. Each cycle shall consist of energizing the control valve with 28 ±1 VDC for 7.5 seconds and deenergizing for 2.5 seconds. The flow rate shall be 28.3 (+10,-0) liters per minute at an inlet pressure of 3000 ± 100 psig. The valve shall be at each of the temperature extremes for 1000 ±50 cycles during the 5000 cycle life test. At the conclusion of the life test the control valve shall be inspected in accordance with 4.6.3 and 4.6.5.

4.6.10 Burst pressure. The control valve shall be subjected to a burst pressure test. The control valve shall be closed, but the outlet fitting shall remain uncapped. The inlet probe shall be hydrostatically pressurized to 6700 ±100 psig for a period of three minutes. There shall be no evidence of permanent deformation.

4.6.11 Shock. The control valve shall be subjected to the shock test in accordance with MIL-STD-810, Method 516, Procedure I, Figure 516-2, except the acceleration shall be 25 g's and shall reach peak value in not more than 0.6 second and not less than 0.015 second. Pressurization is not required. The valve shall be securely mounted to a high strength block. At the completion of the test, the valve shall be visually inspected for damage and shall meet the functional tests of 4.6.3.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Protective caps. Protective caps shall be installed to protect threaded members.

5.1.2 Level C. Each control valve inlet probe shall be protected with a cap free from all contaminants and conforming to MS20813. Control valves shall be individually packaged in a heat-sealable bag conforming to MIL-B-22020, Type I. The control valve shall be packaged free of oil, water and water vapor and the bag shall be dust-free, lint-free and free from water vapor when sealed.

5.2 Packing.

5.2.1 Level C. Control valves shall be packed in accordance with MIL-STD-794.

5.3 Marking. All unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 Intended use. The control valve is a solenoid-operated, normally closed valve. The solenoid is intended to be used in a 28 VDC aircraft system to open the control valve to allow flow of high pressure nitrogen gas.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Government-approved test laboratory (see 3.1.1).

c. First article requirement (see 3.1 and 3.1.1).

d. Option for removal from AS/QVPL (see 3.2.4.3).

e. Option for quality reverification (see 3.2.3).

f. Responsibility for inspection (see 4.1).

g. Samples subjected to sampling inspection (see 4.5.2) are not considered or accepted as part of the contract or purchase order.

h. Rework and retest requirements and quantity to be submitted for inspection lot samples (see 4.5.2).

i. Tests deleted and altered test sequence (see 4.4, 4.5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirement title	Applicable DID no.
3.2	Test Reports	DI-T-21466A
3.2.1, 3.2.2 3.2.4, and 3.2.4.2	Approved Parts Source	UDI-R-20331B
3.2.4.1	Test Procedures	DI-T-3702A
3.3	Engineering Change Proposal	DI-E-2037
4.4.1	Report, First Article Test	UDI-T-23790

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD-5000.19L., Vol. II, AMSL. Copies of data item descriptions required by the contractors in connection

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with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Quality assurance definitions. Definitions for quality assurance terms are in accordance with MIL-STD-109.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item. The first article sample should consist of four control valves. The contracting officer should include specific instructions in all acquisition documents regarding arrangement for examinations, test and approval of the first article.

6.5 Contract or purchase order provision. Contract or purchase orders should specify the following provisions for first article inspection. "When a contractor is in continuous production of the control valve applicable to this specification from contract to contract, consideration should be given to waive the first article inspections. However, waiver of first article inspections should not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract." The words "purchase order" should be substituted throughout for the word "contract" when applicable.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-V-23254C(AS)	Valve, Control, Specification for

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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